

Advanced (Habanero)

Q1. Solve the inequality $|x - 2| < 3$.

- (A) $-1 < x < 5$
 (B) $-4 < x < 1$
 (C) $1 < x < 5$
 (D) $0 < x < 4$
 (E) $2 < x < 6$

Q2. The solution set of $|x| > 4$ is

- (A) $(-\infty, -4) \cup (4, \infty)$
 (B) $(-4, 4)$
 (C) $(-\infty, 4)$
 (D) $(-4, 0) \cup (0, 4)$
 (E) $(-\infty, 0) \cup (0, \infty)$

Q3. The inequality $|x + 3| \leq 2$ can be written as

- (A) $-1 \leq x \leq 5$
 (B) $-5 \leq x \leq -1$
 (C) $1 \leq x \leq 5$
 (D) $-3 \leq x \leq 1$
 (E) $-2 \leq x \leq 4$

Q4. The absolute value inequality $|2x - 1| > 5$ is equivalent to

- (A) $x < -2$ or $x > 3$
 (B) $x < 2$ or $x > 1$
 (C) $x < 1$ or $x > 2$
 (D) $x < -3$ or $x > 1$
 (E) $x < 0$ or $x > 4$

Q5. The solution set of $|x^2 - 4| < 5$ is

- (A) $(-1, 3)$
 (B) $(-3, 1)$
 (C) $(-2, 2)$

(D) $(-\infty, -2) \cup (-1, 3)$

(E) $(-3, -1) \cup (1, 3)$

Q6. The absolute value inequality $|3x - 2| \leq 1$ is equivalent to

- (A) $\frac{1}{3} \leq x \leq 1$
 (A) $\frac{2}{3} \leq x \leq \frac{4}{3}$
 (A) $\frac{1}{3} \leq x \leq \frac{5}{3}$
 (A) $\frac{2}{3} \leq x \leq \frac{4}{3}$
 (A) $\frac{1}{3} \leq x \leq \frac{4}{3}$

Q7. The inequality $|x| - 2 > 1$ is equivalent to

- (A) $x > 3$ or $x < -3$
 (B) $x > 2$ or $x < 1$
 (C) $x > 1$ or $x < -1$
 (D) $x > 1$ or $x < 2$
 (E) $x > 0$ or $x < 4$

Q8. Solve $|2x + 1| < 7$.

- (A) $-2 < x < 3$
 (B) $-4 < x < 1$
 (C) $-3 < x < 2$
 (D) $-1 < x < 4$
 (E) $0 < x < 5$

Open Ended Questions (FRQ)

Q9. Solve $|x^2 - 9| > 3$ and express the solution set in interval notation.

Q10. Model the problem of the temperature range $|T - 50| \leq 10$ where T is in degrees Fahrenheit, and graph the solution set on a number line.

Chef's Tips

- Tip 1: Review the properties of absolute value.
- Tip 2: Emphasize understanding of inequalities and their application.
- Tip 3: Remind students to check their work.